Yuyao Wang

Email: yuyao_wang1@brown.edu | Website: https://wangyuyao98.github.io

Academic Appointments

Postdoctoral Research Associate 2026 - present
Department of Biostatistics, Brown University. Mentor: Larry Han

Visiting Postdoctoral Fellow 2026 - present
Department of Health Care Policy, Harvard Medical School. Mentors: José R. Zubizarreta, Alex Luedtke

Postdoctoral Research Fellow 2025 - 2026
Department of Public Health and Health Sciences, Northeastern University. Mentor: Larry Han

Education

Ph.D. in Mathematics with Specialization in Statistics 2019 - 2025
University of California San Diego. Advisor: Ronghui (Lily) Xu

M.A. in Mathematics 2019 - 2022
University of California San Diego

B.S. in Mathematics 2015 - 2019
Xi'an Jiaotong University

Fellowships

Halicioglu Data Science Institute Graduate Prize Fellowship, UC San Diego, 2019 - 2023.

Awards

NSF Travel Award for the ICSA 2026 Applied Statistics Symposium, 2026.
Silver Reviewer for ICML 2026.
New England Rare Disease Statistics (NERDS) Workshop Best Poster Award, 2025.
UCSD GPSA Travel Grant Award, 2024, 2025.
Society for Causal Inference (SCI) Travel Scholarship, 2024.
Student Paper Competition Award, Lifetime Data Science Conference (LiDS), 2023.

Publications

1. **Yuyao Wang**, Alexander W. Levis, Shu Yang, Larry Han (2026). History-aware conformal prediction sets for censored time-to-event outcomes. *arXiv:2605.06581.* Accepted at *NeurIPS*.
2. **Yuyao Wang**, Andrew Ying, Ronghui Xu (2024). Doubly robust estimation under covariate-induced dependent left truncation. *Biometrika*, 111(3), 789-808.
   (This paper won the Student Paper Competition Award at 2023 Lifetime Data Science Conference)
3. Peng, Y., **Wang, Y.**, Xu, R. (2023). Measures of explained variation under the mixture cure model for survival data. *Statistics in Medicine*, 42(3), 228-245.

ArXiv Preprints

1. **Yuyao Wang**, Andrew Ying, Ronghui Xu (2025). Proximal survival analysis for dependent left truncation. *arXiv:2512.21283. (Major revision at Biometrics)*
2. **Yuyao Wang**, Andrew Ying, Ronghui Xu (2024). A liberating framework from truncation and censoring, with application to learning treatment effects. *arXiv:2411.18879.*
3. Jiyue Qin, **Yuyao Wang**, Ronghui Xu (2026). Doubly Robust Estimation under Covariate Dependent Censoring in Semi-Competing Risks Data. *arXiv:2609.32311.*

Software

R packages:

1. `truncAIPW`: Doubly Robust Estimation under Covariate-Induced Dependent Left Truncation
2. `truncProxy`: Proximal Weighting Estimation for Dependent Left Truncation
3. `aftR2`: R-Squared Measure under Accelerated Failure Time (AFT) Models

Presentations

Talks

History-Aware Conformal Prediction Sets for Censored Time-to-Event Outcomes.

- Joint Statistical Meetings, 2026
- IMS New Researcher Conference, 2026 (lightning talk)

A Liberating Framework from Truncation and Censoring, with Application to Learning Treatment Effects.

- ICSA Applied Statistics Symposium, 2026

Proximal Survival Analysis for Dependent Left Truncation.

- Joint Statistical Meetings, 2025

Learning Treatment Effects under Covariate Dependent Left Truncation and Right Censoring.

- Lifetime Data Science Conference, 2025
- Online Causal Inference Seminar (OCIS), November 5, 2024
- Biostatistics Seminar, Department of Population Medicine, Havard Pilgrim Health Care Institute, 2024
- Causal Inference Seminar, Boston University, 2024
- Causal Group Seminar, Carnegie Mellon University, 2024
- Joint Statistical Meetings, 2024
- Southern California Applied Mathematics Symposium, 2024

Doubly Robust Estimation under Covariate-induced Dependent Left Truncation

- McGill Statistics Seminar, 2023
- Joint Statistical Meetings, 2023
- Lifetime Data Science Conference, 2023

Posters

History-Aware Conformal Prediction Sets for Censored Time-to-Event Outcomes.

- STAI-X: Statistics and Trustworthy AI for Cross (X)-Domain Acceleration, 2026
- IMSI workshop: New Horizons on Model Transportability and Data Integration, 2026
- American Causal Inference Conference, 2026

Learning Treatment Effects under Covariate Dependent Left Truncation and Right Censoring.

- New England Rare Disease Statistics (NERDS) Workshop, 2025
- Public Health Research Day at UCSD, 2025

Doubly Robust Estimation of Treatment Effects under Covariate Dependent Left Truncation and Right Censoring.

- American Causal Inference Conference, 2024
- Public Health Research Day at UCSD, 2024

Multiply Robust Estimation of Treatment Effect for Time-to-event Outcome under Dependent Left Truncation.

- American Causal Inference Conference, 2023
- Public Health Research Day at UCSD, 2023

Semiparametric Estimation for Non-randomly Truncated Data.

- American Causal Inference Conference, 2022
- Public Health Research Day at UCSD, 2022

Research Experience

Biostatistics internship June – August 2024

St. Jude Children's Research Hospital, Memphis, TN | Mentor: Kendrick Li

- Developed inverse probability weighting approaches for handling the selection bias involved in the SJLIFE cancer survivorship study under semi-competing risks settings.
- Conducted simulation studies and analyzed the cardiotoxicity of childhood cancer treatments using SJLIFE data.
- Derived the efficient influence curve for a general Z-estimand under semi-competing risks settings with covariate dependent left truncation.

Research Assistant October 2023 – June 2024

University of California San Diego, La Jolla, CA | Supervisor: Ronghui Xu

- Analyzed factors that protect against the development of Alzheimer's Disease neuropathological changes using brain autopsy data from Honolulu Asia Aging Study.
- Applied inverse probability weighting approaches for handling the selection bias from double truncation.

Research Assistant 2023 - 2024

UCSD-based Organization of Teratology Information Services research center | Supervisor: Ronghui Xu

- Helped with the statistical analysis plan for assessing the safety of medications during pregnancy; particularly in addressing challenges in analyzing the effect of different timings of exposure in the presence of the selection bias due to left truncation.

Teaching Experience

Teaching Assistant 2019 - 2025

University of California San Diego, La Jolla, CA

- Statistics and biostatistics courses: Lifetime Data Analysis; Applied Survival Analysis; Introduction to Mathematical Statistics; Introduction to Computational Statistics; Data Analysis and Inference; Calculus-Based Probability and Statistics
- Data science courses: Data Science Project (capstone)
- Math courses: Calculus

Service

Peer Review Service:

- Journals: Journal of the Royal Statistical Society: Series B, Biometrics, Statistics in Medicine, International Journal of Biostatistics, Journal of Causal Inference, Lifetime Data Analysis
- Conferences: ICML, NeurIPS

Conference organizing:

- Co-organizer and chair of the topic-contributed session "Balancing and Automatic Debiased ML Methods for Causal Inference" at 2026 Joint Statistical Meetings.
- Co-organizer and chair of the invited session "Recent Advances in Causal Inference to Address Complexities in Real-world Data" at 2025 Joint Statistical Meetings.
- Organizer of the topic-contributed paper session "Recent Advances in Biased Sampling, Generalizability and Transportability" at 2025 Joint Statistical Meetings.
- Chair of the invited session "Recent Advances in Truncated and Censored Data" at 2025 Lifetime Data Science Conference.
- Member of the Scientific Organizing committee for 2025 International Conference on Health Policy Statistics.
- Co-organizer of the topic-contributed paper session "Causal Inference with Complex Missing Patterns" at 2024 Joint Statistical Meetings.

Other service:

- Harvard SPIDEY reading group coordinator, 2026.

- UCSD Association of Women in Math (AWM) Colloquium coordinator, 2024.
- Mentor in the Math Graduate Mentorship Program at UC San Diego, 2020, 2021, 2024.
- Mentor in the High School Math Program at UC San Diego, Summer 2023.